

## Solution Requirements

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 15 March 2026                   |
| <b>Team ID</b>       | SWTID-2026-5556                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 4 Marks                         |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                     | Priority (High/Medium/Low) |
|------|----------------------|-------------------------------------------------------------|----------------------------|
| 1    | User Registration    | User can register and create an account.                    | High                       |
| 2    | User Login           | User can securely log in using username/email and password. | High                       |

|   |                         |                                                                                  |        |
|---|-------------------------|----------------------------------------------------------------------------------|--------|
| 3 | Credit Card Application | User can enter personal and financial details to apply for a credit card.        | High   |
| 4 | Data Validation         | System validates all required input fields before submission.                    | High   |
| 5 | ML Prediction           | Machine Learning model predicts whether the application is Approved or Rejected. | High   |
| 6 | Result Display          | Display approval status instantly to the user.                                   | High   |
| 7 | Admin Dashboard         | Admin Dashboard                                                                  | Medium |
| 8 | Report Generation       | Generate application and prediction reports.                                     | Medium |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category               | Requirement Description              | Target Metric / Acceptance Criteria |
|------|----------------------------|--------------------------------------|-------------------------------------|
| 1    | Performance & Speed        | Prediction should be fast            | Response time < 3 seconds           |
| 2    | Scalability                | Security                             | Protect user information.           |
| 3    | Security & Data Privacy    | System should work without failures. | 99% availability                    |
| 4    | Reliability & Availability | Easy-to-use interface.               | Simple and responsive UI            |

|          |                           |                             |                       |
|----------|---------------------------|-----------------------------|-----------------------|
| <b>5</b> | Usability & Accessibility | Support multiple users.     | 100+ concurrent users |
| <b>6</b> | Maintainability           | Easy to update and maintain | Modular Python code   |